

Haohan Yuan



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SUMMARY

PhD researcher in Computer Science at the University of Hawaii at Manoa, working on reliable and explainable large language model systems. My research focuses on LLM reasoning, long-context understanding, structural priors, and systematic evaluation, with publications at ACL, ICML, NAACL, and EACL. I am actively seeking research internship or visiting student opportunities in LLM algorithms, multimodal reasoning, agentic AI, and reliable AI systems.

RESEARCH INTERESTS

Reliable and explainable AI systems; structural priors for LLM/VLM reasoning; long-context understanding; multimodal and agentic AI; benchmark design and systematic evaluation.

EDUCATION

University of Hawaii at Manoa

▪ *Ph.D. Student in Computer Science, Information and Computer Sciences Department*

◦ **Advisor:** Haopeng Zhang

Hawaii, USA

Aug. 2024–Present

Nanyang Technological University

▪ *M.Eng. in Computer Science, College of Computing and Data Science*

◦ **Advisor:** Siu Cheung Hui

Singapore

Aug. 2022–Jul. 2024

Hefei University of Technology

▪ *B.Sc. in Mathematics and Applied Mathematics, School of Mathematics*

Hefei, China

Aug. 2018–May 2022

SELECTED PUBLICATIONS

(* denotes equal contribution)

- [C1] **Haohan Yuan**, Haopeng Zhang, “Understanding LLM Reasoning for Abstractive Summarization”. In *Findings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 (Top-tier NLP Conference))* Studies how explicit reasoning affects summarization quality and factual faithfulness.
- [C2] Yizhu Wen, Nan Zhang, **Haohan Yuan**, Xun Chen, Haopeng Zhang, Hanqing Guo, “Position: Generative Engine Optimization Creates Underexamined Risks, Governance Must Target Concentration, Disclosure, and Academic Blind Spots”. In *The 43rd International Conference on Machine Learning (ICML 2026 (Position Track, Top-tier ML Conference))* Identifies risks involving concentrated influence, disclosure, and evaluation blind spots in generative engine optimization.
- [C3] **Haohan Yuan**, Sukhwa Hong, Haopeng Zhang, “StrucSum: Graph-Structured Reasoning for Long Document Extractive Summarization with LLMs”. In *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics. (EACL 2026 (Top-tier NLP Conference))* Proposes graph-structured reasoning for long-document extractive summarization.
- [C4] **Haohan Yuan**, Haopeng Zhang, “DomainSum: A Hierarchical Benchmark for Fine-Grained Domain Shift in Abstractive Text Summarization”. In *Findings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics. (NAACL 2025 (Top-tier NLP Conference))* Introduces a benchmark for summarization robustness under fine-grained domain shifts.
- [C5] **Haohan Yuan**, Siu Cheung Hui, Haopeng Zhang, “A Structure-aware Generative Model for Biomedical Event Extraction”. In *The 30th International Conference on Database Systems for Advanced Applications (DASFAA 2025)* Proposes structure-aware generation for biomedical event extraction.

SELECTED RESEARCH PROJECTS

StrucVLA: Structural Priors for Explainable Vision-Language-Action Policies

- *Ongoing Research*
 - Developing a structured high-level task state representation for embodied policies, decomposing implicit VLA understanding into scene, task, interaction, and progress variables.
 - Building a Unified State Binding pipeline that uses multimodal LLMs as offline semantic teachers to generate structured state supervision with verifier- and rule-based quality control.
 - Evaluating high-level state representations through field-level diagnosis, block ablations, corrupted-state tests, and counterfactual state control.

SumFingerprint: Reliability and Fragility of LLM Summary Fingerprints

▪ Ongoing Research

- Studying whether LLM-generated summaries contain model-specific stylistic fingerprints that remain reliable under rewriting and transformation attacks; built controlled attribution benchmarks and interpretable feature analyses, with preliminary evidence that such fingerprints are observable under controlled settings.
- Analyzing fingerprint fragility and targeted misattribution risks, motivating more robust evaluation of model provenance in high-stakes summarization systems.

Structured Reasoning and Evaluation for LLM Summarization

▪ Published Research Thread

- Developed graph-structured reasoning methods for long-document summarization and studied domain shift, explicit reasoning, and factual faithfulness, connecting structured reasoning with robustness evaluation and reliable text generation.

PROFESSIONAL EXPERIENCE

SPIRIT Laboratory, Nanyang Technological University / Singapore Food Agency

Singapore

▪ Software Engineering Intern

Oct. 2022–Apr. 2023

- Built a full-stack multilingual food safety news monitoring platform with Django, Vue.js, and Elasticsearch, supporting news ingestion, indexing, search, and risk tracking.
- Implemented retrieval, filtering, dashboard, and keyword-based alerting workflows for food safety event monitoring by domain analysts.

iFLYTEK CO., LTD.

Hefei, China

▪ Machine Learning Intern

Sep. 2021–Dec. 2021

- Developed a BERT-based named entity recognition pipeline to extract structured product attributes from noisy e-commerce advertisement text.
- Built data preprocessing and annotation workflows for noisy commercial text to support downstream attribute extraction.

HONORS AND AWARDS

Hawaii Data Science Institute AI @ HIDSII Fellowship

2025

PROFESSIONAL SERVICE

Conference Reviewer and Program Committee Member

- ACL 2026, AAAI 2026, NewSumm 2025 (EMNLP Workshop)

Conference Volunteer

- NAACL 2025

SKILLS

Programming

- Python, JavaScript, SQL, MATLAB

Machine Learning

- PyTorch, Hugging Face Transformers, model fine-tuning, LoRA, prompt engineering, benchmark evaluation

LLM & NLP

- LLM reasoning, multimodal LLMs/VLMs, long-context, summarization, information extraction/NER, domain shift evaluation

Systems & Tools

- vLLM, Django, Vue.js, Elasticsearch, OpenAI API, Git, Linux